

01NAEX - Lecture 00

General information on the course

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Basic information

Duration of Course:	13 weeks.
Schedule:	Every Monday 15:00 to 17:30, room T-112.
Materials:	https://github.com/francji1/01NAEX
Type of Assessment:	exercises, two team projects, and final report.
Evaluation:	6 step scale (A,B,C,D,E, F).
Qualified Prerequisites:	Regression Analysis, Statistics Experience with programming in R or in Python.
Scope and Form:	lectures, discussion of exercises in lecture room, exercises in groups, team projects, reports.
Guarantee:	J. Franc & T. Hobza
Teaching:	J. Franc

Goal of the 01NAEX course

- ▶ The main aim of this course is to learn how to design and analyse an experiment where variation is present. Given experimental conditions, the main challenge is to formulate experimental plans which will provide informative data suitable for statistical analysis.
- ▶ Secondary goal is to learn how to build a tiny project from beginning (measure data) to final presentation of the analysis, improve your data analysis skills and think about data processing in many aspects.

Outline of the course (can be changed :)

1. Introduction and simple comparative experiments
2. Single factor experiment ANOVA
3. Factorial design
4. Blocking, Latin squares, Graeco-Latin squares, BIBD
5. 2^k factorial design I
6. 2^k factorial design II
7. 2^k factorial design - Blocking and Center point
8. 2^k fractional factorial design
9. 3^k factorial design
10. Random factors mixed models
11. Nested and split plot
12. Longitudinal Data Analysis
13. Final Project

Assessment and Evaluation details

The system of assessment and evaluation of the 01NAEX course is the following:

Final grade is primary based on **Final written report** and examination based on discussion of the project (scheduled dates during the exam period).

Attend the exercises and hand in the assignment from the end of each exercise (Python Jupyter notebook, pull request to the course GitHub repository, folder HW/) before the next lecture. If you miss the lecture, elaborate and hand in the exercise at home. Two missed lectures are tolerated without penalization.

Not satisfying the condition $\Rightarrow$ -1 grade for each miss.

Elaborate and hand in **two team projects** during the semester in **satisfactory quality**.

Not satisfying the condition $\Rightarrow$ -1 grade for each project.

Literature

Main textbook:

- ▶ **D. C. Montgomery (2012): Design and Analysis of Experiments, 8th Edition, John Wiley and Sons.**

Exercise and problem numbers used in the course follow this edition.

https://faculty.ksu.edu.sa/sites/default/files/douglas_c._montgomery-design_and_analysis_of_experiments-wiley_2012_edition_8.pdf

Secondary textbook:

- ▶ A. Dean, D. Voss, D. Draguljić (2017): Design and Analysis of Experiments, 2nd Edition, Springer.

<https://link.springer.com/book/10.1007/978-3-319-52250-0>